IDS- Theory-2.

Q1 Characteristics are:
1. Contiguous memory location
2. Indexed Access
3. Fixed size
4. Fast Random Access
5. Insertion & deletion are costly

Memory Wastage Possibility
Simple Implementation
Ordered Arrangement.

2. An array is a fundamental abstract data Type that models an ordered, fixed - size collection of elements, all of the same type, which can be accessed efficiently by their index or position.

Components of ADT:
Elements: Data items of the same type
Index: Position used to access elements
Size: Total no of elements.

Basic operation on array
Traversal: visiting each element
Insertion: adding a new element
Selection: Removing an element
Searching: Finding an element.

3. Consider an array A = [10, 20, 30, 40]

Insertion: Insert 25 at position 2:
Shift elements right → A = [10, 20, 25, 30]

Deletion: Delete element at position 3:
Shift elements left →
A = [10, 20, 25, 40]

Searching: To search 25.
Compare sequentially until found
25 is found at index 2.

4. Advantages: Fast direct acess using index
Simple & easy to implement. Efficiency memory usage for fixed-sized Good for storing homogeneous data.

limitation:
Fixed size
Insertion & deletion are time-consuming
wastage at memory if size not used.

efficiency & comparision
Insertion: O (n) due to Shifting
Deletion: O (n) due to Shifting
Searching: O (n) for linear seach, O (log n) fo binary seach (sorted array).

3. Efficient manipulation:
Direct Indexing allows fast access.
Nested loops help in processing ~~total~~ & average.

Extension to n-dimensional array
- Image processing
- Scientific stimulation
- ML & data analytics
- Rov development.